

Unit-IV: Regularization for Deep Learning

❖ Overview of Overfitting

Overfitting is a fundamental problem in Machine Learning and Deep Learning where a model learns the **training data too well**, including its noise and irrelevant patterns, and therefore performs poorly on new, unseen data.

Causes of Overfitting

(a) Complex Models

- Too many parameters (e.g., deep neural networks)
- Model capacity > data complexity

(b) Small Dataset

- Insufficient data leads to memorization

(c) Noisy Data

- Model learns random fluctuations as patterns

(d) Too Many Features

- Irrelevant or redundant features confuse the model

(e) Long Training Time

- Model keeps adjusting until it fits noise

Symptoms of Overfitting

You can identify overfitting when:

- Training loss ↓ continuously
- Validation loss ↓ initially, then ↑
- Large gap between training and validation accuracy

Example

Imagine training a model to recognize cats:

- Training images: 100 cats
- Model memorizes:
 - Background
 - Lighting
 - Exact pixel patterns

☞ When shown a new cat image → fails

Because it learned **specific examples**, not the concept of a “cat”.

Real-World Analogy

Think of a student:

- **Overfitting student:**
 - Memorizes answers word-by-word
 - Fails when questions change slightly
- **Well-trained student:**
 - Understands concepts
 - Can solve new problems

Effects of Overfitting

- Poor generalization
- Unreliable predictions
- Model fails in real-world deployment

Techniques to Prevent Overfitting

- Regularization (L1, L2)
- Dropout
- Early stopping
- Data augmentation
- Reduce model complexity
- More training data

❖ Types of Biases

In Deep Learning, “bias” can refer to **multiple concepts**—from mathematical errors in models to data-related and human-induced biases. Understanding these is essential for designing robust and fair systems.

1. Model (Statistical) Bias

Definition

Model bias is the **error caused by simplifying assumptions** in a neural network.

Key Idea

- High bias → Model is too simple
- Leads to **underfitting**

Example

- Using a shallow neural network for a highly complex task like image recognition

Characteristics

- Poor performance on both training and test data
- Cannot capture complex patterns

2. Variance (Closely Related Concept)

Definition

Variance refers to how much the model output changes with different training data.

Key Idea

- High variance → Overfitting
- Model memorizes training data

Characteristics

- Low training error
- High validation/test error

3. Data Bias

Definition

Bias introduced due to **issues in training data**, very common in deep learning.

Types

(a) Sampling Bias

- Dataset not representative
- Example: Face recognition trained mostly on one ethnicity

(b) Class Imbalance

- Some classes dominate
- Example: 95% normal, 5% anomaly

(c) Label Bias

- Incorrect or inconsistent labeling

(d) Measurement Bias

- Faulty sensors or poor data collection

4. Algorithmic Bias

Definition

Bias introduced due to **model architecture or training process**.

Examples

- CNNs assume spatial locality
- RNNs assume sequential dependencies

Causes

- Initialization
- Optimization algorithms
- Loss functions

5. Inductive Bias

Definition

The **assumptions that help neural networks generalize** from limited data.

Examples

- Convolutional Neural Networks (CNNs) assume:
 - Local connectivity
 - Translation invariance

Importance

- Essential for learning
- Without it, models cannot generalize

6. Representation Bias

Definition

Occurs when the model fails to **capture important features** of the data.

Example

- A model ignoring rare but important features

Impact

- Poor feature learning
- Weak generalization

7. Optimization Bias

Definition

Bias introduced during the **training process (optimization)**.

Causes

- Gradient descent getting stuck in local minima
- Poor learning rate selection

Impact

- Suboptimal model performance

8. Cognitive (Human) Bias

Definition

Bias introduced by **human decisions** in the pipeline.

Examples

- Choosing biased datasets
- Designing models based on assumptions
- Selective evaluation

9. Evaluation Bias

Definition

Bias due to **incorrect evaluation methods**.

Examples

- Using training data for testing
- Wrong metrics (e.g., accuracy for imbalanced data)

10. Historical Bias

Definition

Bias arising from **historical data reflecting real-world inequalities**.

Example

- Hiring AI trained on biased past hiring decisions

11. Deployment Bias

Definition

Occurs when the model is used in a **different environment than it was trained for**.

Example

- Model trained on high-quality images fails on low-resolution real-world data

❖ Bias–Variance Tradeoff

In Deep Learning and Machine Learning, model performance depends on balancing two types of errors:

- **Bias** → Error due to overly simple assumptions
- **Variance** → Error due to sensitivity to training data

Intuition

- **High Bias (Underfitting)**
 - Model too simple
 - Misses important patterns
- **High Variance (Overfitting)**
 - Model too complex
 - Learns noise

Visual Understanding

☞ As model complexity increases:

- Bias ↓
- Variance ↑

☞ Goal: **Find optimal balance** where total error is minimum.

Mathematical Insight

Total Error can be approximated as:

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

- **Bias²** → Error from assumptions
- **Variance** → Error from fluctuations
- **Irreducible error** → Noise in data (cannot be removed)

❖ Role of Regularization

What is Regularization?

Regularization is a technique used to **reduce overfitting (high variance)** by adding a **penalty** to the model complexity.

☞ It discourages large weights in neural networks.

L1 Regularization (Lasso)

Definition

L1 regularization adds the **absolute value of weights** as a penalty.

Key Characteristics

- Produces **sparse models** (many weights become zero)
- Performs **feature selection automatically**
- Useful when many features are irrelevant

Example

In a dataset with 100 features:

- L1 may reduce it to **10 important features**
- Remaining weights → zero

Advantages

- Feature selection
- Simple and interpretable models

Disadvantages

- Can be unstable when features are highly correlated

L2 Regularization (Ridge)

Definition

L2 regularization adds the **square of weights** as a penalty.

Key Characteristics

- Does **not make weights zero**, but makes them small
- Distributes importance across all features

Example

Instead of removing features:

- L2 assigns **small weights** to less important features

Advantages

- Stable solution
- Works well with correlated features

Disadvantages

- Does not perform feature selection

❖ Parameter Sharing

Parameter sharing is a key concept in Deep Learning that allows a model to **reuse the same parameters (weights)** across different parts of the input. It is widely used to make models **efficient, scalable, and better at generalization**.

What is Parameter Sharing?

Definition

Parameter sharing means using the **same set of weights multiple times** in a model instead of learning separate weights for every input feature.

☞ Instead of learning many different parameters, the model **shares one set across multiple computations**.

Simple Intuition

Think of scanning a document with the same highlighter:

- You don't use a different highlighter for each word
- You use the **same highlighter everywhere**

☞ Similarly, neural networks reuse the same weights.

Why Parameter Sharing is Needed

Without parameter sharing:

- Number of parameters becomes **very large**
- Model becomes:
 - Slow
 - Memory-intensive
 - Prone to **overfitting**

With parameter sharing:

- Fewer parameters
- Faster training
- Better generalization

Parameter Sharing in Convolutional Neural Networks (CNNs)

Key Idea

In **CNNs**, the same filter (kernel) is applied across the entire image.

Visual Understanding

- A small filter (e.g., 3×3) scans the image
- The **same weights** are used at every position
- This produces a **feature map**

Example

Detecting edges in an image:

- Instead of learning separate edge detectors for each region
- CNN uses **one filter** that works everywhere

Benefits in CNNs

- Reduces parameters drastically
- Captures **spatial patterns**
- Enables **translation invariance**

Parameter Sharing in Recurrent Neural Networks (RNNs)

Key Idea

In **RNNs**, the same weights are reused across time steps.

Visual Understanding

- Each time step uses the **same parameters**
- Helps process sequences like:
 - Text
 - Speech
 - Time series

Example

Sentence processing:

- Same network processes each word
- Learns patterns like grammar and context

Advantages of Parameter Sharing

(a) Reduces Number of Parameters

- Smaller model size
- Less memory required

(b) Prevents Overfitting

- Fewer parameters → less chance of memorization

(c) Improves Generalization

- Learns patterns that apply across different locations

(d) Efficient Learning

- Faster training
- Requires less data

Disadvantages of Parameter Sharing

(a) Limited Flexibility

- Same weights everywhere → may miss location-specific patterns

(b) Not Suitable for All Tasks

- Tasks requiring unique features at each position may not benefit

❖ Dropout

Definition

Dropout randomly “turns off” (drops) a fraction of neurons during training.

How it Works

- During training: randomly remove neurons with probability p
- During testing: use full network (with scaled weights)

Intuition

- Prevents neurons from **co-adapting**
- Forces network to learn **robust features**

Advantages

- Reduces overfitting
- Acts like training multiple models (ensemble effect)

Disadvantages

- Slower convergence
- Requires tuning dropout rate

❖ Weight Decay (L2 Regularization)

Definition

Adds a penalty on large weights to the loss function.

Key Idea

- Penalizes large weights
- Keeps weights **small and smooth**

Intuition

- Large weights → complex model → overfitting
- Weight decay → simpler model

Advantages

- Improves generalization
- Stable training

❖ Batch Normalization

Definition

Batch Normalization (BatchNorm) normalizes the input of each layer to have:

- Mean ≈ 0
- Variance ≈ 1

Key Idea

- Reduces **internal covariate shift**
- Stabilizes learning

Advantages

- Faster training
- Allows higher learning rates
- Acts as slight regularization

Disadvantages

- Adds complexity
- Depends on batch size

❖ Early Stopping

Definition

Stop training when validation performance starts to degrade.

How it Works

- Monitor validation loss
- Stop when:
 - Validation loss increases
 - Training loss continues decreasing

Intuition

- Prevents model from learning noise

Advantages

- Simple and effective
- Saves training time

❖ Data Augmentation

Definition

Artificially increases dataset size by **transforming existing data**.

Techniques

For Images:

- Rotation
- Flipping
- Cropping
- Scaling
- Color jitter

For Text:

- Synonym replacement
- Paraphrasing

Intuition

- Exposes model to **varied data**
- Improves generalization

Advantages

- Reduces overfitting
- No need for new data collection

❖ Adding Noise to Input and Output

Definition

Introduce random noise into:

- Input data
- Output labels (sometimes)

Visual Understanding

Types of Noise

(a) Input Noise

- Add Gaussian noise to inputs
- Makes model robust to variations

(b) Output Noise (Label Noise)

- Slightly perturb labels
- Prevents overconfidence

Intuition

- Forces model to learn **general patterns**
- Not memorize exact values

Advantages

- Improves robustness
- Acts like regularization

***** All The Best *****